

Many Good Things, With An Open Twist

Gone are the days when 'open source' was a concept that applied mainly to the arts and software. Today, open source fires innovation in many things, from gadgets and cars to prosthetic limbs and advanced military vehicles. Every month, in this section, we take a peek at the open source aspects of some interesting devices, applications, tools, etc.

Truly the car of the future

Visteon Corporation has developed the e-Bee, a concept car based on the Nissan Leaf. It has all the automotive electronics you would like to see in your future car – and if Visteon's estimate is right, your dream might come true by 2020. The e-Bee has everything from entertainment and mobility solutions to safety and comfort. It is filled with loads of technology, including a personalised human-machine interface (HMI), touch-screen instrument panels that give information on vehicle controls and social media connections, a head-down projected main display in front of the driver, a rear view with 360-degree visibility and augmented reality features, tech-agnostic wireless charging, near-field communications (NFC) for a personal link between user and vehicle, climate control system, cloud connectivity and car-to-car communications to improve driver information about collisions, hazardous roads, curve speed warnings and traffic flow, and much more.



The open twist: e-Bee's extremely advanced audio, video and information platform is based on open technology, and is designed to be upgraded automatically throughout the car's lifetime. Using open architecture and open source software that meet GENIVI standards, this solution can operate as both the HMI and Media-Oriented Systems Transport (MOST) master and provide up to four independent instances of HMI. The system integrates a wide range of functions into a single scalable module while providing varied connectivity features from Bluetooth and Wi-Fi to Internet connectivity, remote HMI configuration, and software and feature updates. The open software solution leverages features developed by several open source software GENIVI members, maximising re-use and robustness, while reducing the engineering costs to develop such complex systems.

Micro-clouds for a 'connected' classroom

One of the best devices demonstrated at CES 2013 was Marvell's SMILE Plug, which is an easy-to-use, low-power cloud computer that claims to turn a normal classroom into a connected, secure and highly-interactive learning environment by setting up a micro-



cloud for up to 60 students. It facilitates what Marvell calls Classroom 3.0, by enabling the delivery of educational content and applications via a range of mobile devices including smart phones and tablets. The device is powered by Marvell's

high-performance, low-power ARMADA 300 series system-on-a-chip (SoC) and the company's industry leading Avastar 88W8764 wireless chip. Network connectivity is through Gigabit Ethernet, and peripheral devices can be connected using USB 2.0 or Wi-Fi.

The open twist: The SMILE Plug runs on a completely open source platform, which makes it easy to develop or port any additional learning applications. It is based on Arch Linux for ARM, NODE.js, and Stanford's Mobile Inquiry-Based Learning Environment (SMILE) and software development kit (SDK). Other open components include the Node Package Manager, the Pluginmin administration API and UI (that runs on Android-based client devices) and the SMILE Junction Server Administration Client.

An open source prosthetic hand

When Richard Van As of South Africa lost four of his fingers in a sawing accident, he stumbled upon the designs of a prosthetic hand on YouTube. Together, Van As and the original developer Ivan Owen from Washington perfected the designs to suit Van As. Soon, they decided to make this a worldwide campaign to help those who needed prosthetic hands, and uploaded the designs of their body-powered device in the public domain. They continue to work on the designs, and welcome donations, suggestions and improvements. They also received two 3D printers as a donation from Makerbot, which allows them to synthesise their designs. In January, they used these designs to develop a 3D-printed prosthetic hand, which they called Robohand, for a South African five-year old boy called Liam, who was born without fingers on his right hand. The Robohand suits Liam quite well, and he is now able to play, take care of his personal needs and attend school—where he shows off his mechanical fingers!

The open twist: Robohand is a typical example of low-tech mechanics, fast-prototyping and the power of open source. It is made of 16 3D-printed pieces and 28 common, off-the-shelf items like nylon cord, nuts, bolts and rubber thimbles. The prosthetic hand is controlled by a system that reacts to the